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Review

Neuroplasticity Mechanism in Post-Stroke Language Recovery

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ARTICLE INFO

Article history:
Received 02 July 2026
Received in revised form 05 August 2026
Accepted 03 September 2026

Keywords:
Aphasia
Stroke
Neuronal plasticity
Transcranial magnetic stimulation
Diffusion tensor imaging
Language recovery
Neuromodulation
Repetitive transcranial magnetic stimulation

ABSTRACT

Introduction: Recovery from post-stroke aphasia is highly variable. A central clinical dilemma is that existing neuroplasticity models offer conflicting predictions: the vicariation model recommends recruiting contralesional right-hemisphere regions, while the interhemispheric competition model advocates inhibiting them. This disconnect directly impacts neuromodulation decisions.

Objective: This review synthesizes current evidence on neuroplasticity mechanisms in aphasia recovery and proposes a clinically applicable framework to reconcile competing models.

Results: The bimodal balance–recovery model integrates both frameworks by demonstrating that the optimal strategy depends on structural reserve. Patients with preserved ipsilesional white matter integrity (e.g., arcuate fasciculus) benefit from facilitatory ipsilesional stimulation. Those with extensive left-hemisphere damage may require contralesional compensation. Recent meta-analyses indicate that repetitive transcranial magnetic stimulation (rTMS) combined with language therapy produces moderate improvements in naming and repetition (standardized mean difference (SMD) ≈ 0.7). Transcranial direct current stimulation (tDCS) shows small-to-moderate additive effects. Diffusion tensor imaging (DTI)-based assessment of white matter integrity is emerging as a key biomarker for treatment personalization.

Conclusions: No universal recovery mechanism exists. Recovery involves molecular plasticity, structural remodeling, functional network reorganization, and experience-dependent adaptation. The bimodal balance–recovery model currently offers the most clinically useful framework. We recommend integrating neuroimaging biomarkers to guide personalized neuromodulation, combining either ipsilesional facilitation or contralesional compensation with intensive speech-language therapy. This framework is most directly applicable to chronic non-fluent and anomic aphasias; however, further validation is needed for patients with severe global aphasia.

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https://doi.org/10.53986/ibjm.2026.0022

Mecanismo de la neuroplasticidad en la recuperación del lenguaje después de un accidente cerebrovascular

INFO. ARTÍCULO

Historia del artículo:
Recibido 02 Julio 2026
Recibido en forma revisada 05 Agosto 2026
Aceptado 03 Septiembre 2026

Palabras clave:
Afasia
Ictus
Plasticidad neuronal
Estimulación magnética transcraneal
Imagen por tensor de difusión
Recuperación del lenguaje
Neuromodulación
Estimulación magnética transcraneal repetitiva

RESUMEN

Introducción: La recuperación de la afasia post-ictus es muy variable. Un dilema clínico fundamental es que los modelos de neuroplasticidad existentes ofrecen predicciones contradictorias: el modelo de vicariación recomienda activar las regiones contralaterales del hemisferio derecho, mientras que el modelo de competencia interhemisférica aboga por su inhibición. Esta discrepancia influye directamente en las decisiones de neuromodulación.
Objetivo: Esta revisión sintetiza la evidencia actual sobre los mecanismos de neuroplasticidad en la recuperación de la afasia y propone un marco clínicamente aplicable para conciliar los modelos contrapuestos.
Resultados: El modelo bimodal de equilibrio-recuperación integra ambos marcos al demostrar que la estrategia óptima depende de la reserva estructural. Los pacientes con integridad preservada de la sustancia blanca ipsilesional (p. ej., fascículo arqueado) se benefician de la estimulación ipsilesional facilitadora. Aquellos con daño extenso en el hemisferio izquierdo pueden requerir compensación contralesional. Metaanálisis recientes indican que la estimulación magnética transcraneal repetitiva (EMTr) combinada con terapia del lenguaje produce mejoras moderadas en la denominación y la repetición (diferencia de medias estandarizada (DME) ≈ 0,7). La estimulación transcraneal con corriente continua (ETCC) muestra efectos aditivos de leves a moderados. La evaluación de la integridad de la sustancia blanca mediante imágenes de tensor de difusión (ITD) se perfila como un biomarcador clave para la personalización del tratamiento.
Conclusiones: No existe un mecanismo de recuperación universal. La recuperación implica plasticidad molecular, remodelación estructural, reorganización de la red funcional y adaptación dependiente de la experiencia. El modelo bimodal de equilibrio-recuperación ofrece actualmente el marco más útil desde el punto de vista clínico. Recomendamos integrar biomarcadores de neuroimagen para guiar la neuromodulación personalizada, combinando la facilitación ipsilesional o la compensación contralesional con terapia intensiva del habla y el lenguaje. Este marco es más directamente aplicable a las afasias crónicas no fluidas y anómicas; sin embargo, se requiere una mayor validación para pacientes con afasia global grave.

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HOW TO CITE THIS ARTICLE: Rakhmatullaeva GK, Kadirova AS, Botirov SK. Neuroplasticity Mechanism in Post-Stroke Language Recovery. Iberoam J Med. 2026;8(4):140-147. doi: 10.53986/ibjm.2026.0022.

1. INTRODUCTION

Two competing models currently dominate the understanding of post-stroke aphasia recovery. One holds that the right hemisphere can assume language functions. The other contends that right hemisphere activity impedes left hemisphere recovery. These opposing views lead to contradictory treatment strategies — and resolving this tension matters for thousands of patients.

Stroke remains the leading cause of aphasia. Ischaemic stroke accounts for 85–90 % of cases, triggering energy failure, excitotoxicity, calcium influx, and apoptosis following arterial occlusion. Infarct core and penumbra determine recovery potential [1]. Yet lesion size alone does not fully explain the considerable variability in recovery outcomes. Although stroke is more common in older adults, younger patients represent an increasingly important subgroup because of their distinct etiological profile, including hypercoagulable states and uncommon vascular disorders [2].

Aphasia disrupts the dominant hemisphere language network, and recovery trajectories remain notoriously unpredictable [3].

Recovery after post-stroke aphasia is influenced by multiple interacting factors that extend beyond lesion size alone. Age, lesion location, structural connectivity, and cognitive reserve all shape functional reorganization and influence language recovery. Emerging neuroimaging evidence suggests that preservation of white matter pathways, particularly the arcuate fasciculus, may predict recovery potential more accurately than lesion volume alone [4-6].

These findings underscore the need for individualized rehabilitation strategies guided by neurobiological rather than purely anatomical characteristics.

Against this background, this review aims to critically examine current models of post-stroke language recovery, evaluate the available evidence, and propose a biomarker-based strategy for selecting the most appropriate therapeutic approach [3].

2. LITERATURE SEARCH STRATEGY

We conducted a targeted search in PubMed, Scopus, and Google Scholar for articles published between January 2017 and December 2024. The search strategy combined MeSH terms and free-text keywords: ("aphasia" OR "dysphasia") AND ("stroke" OR "cerebrovascular accident") AND ("neuroplasticity" OR "language recovery" OR "reorganisation") AND ("transcranial magnetic stimulation" OR "rTMS" OR "transcranial direct current stimulation" OR "tDCS" OR "DTI" OR "diffusion tensor imaging"). Additional records were identified through reference lists and the "cited by" feature in Google Scholar.

Inclusion criteria were: (1) original research, systematic reviews, meta-analyses, or clinical trials; (2) human participants with post-stroke aphasia; (3) English language; (4) publication in peer-reviewed journals. Exclusion criteria were: (1) animal studies; (2) conference abstracts without full text; (3) case reports with fewer than three participants; (4) studies focused exclusively on non-stroke aetiologies.

The search yielded several hundred records. After removing duplicates and screening titles and abstracts, we reviewed relevant full texts. The final synthesis includes 19 sources that directly address neuroplasticity mechanisms or clinical interventions in post-stroke aphasia recovery. We prioritised recent meta-analyses (2021–2024) and longitudinal neuroimaging studies providing the highest level of evidence.

3. CONTEMPORARY LANGUAGE NETWORK MODEL

Traditional models localized language primarily to Broca's and Wernicke's areas. However, contemporary neuroimaging has demonstrated that language processing depends on a distributed cortical-subcortical network rather than isolated cortical centres. Within this framework, the dual-stream model distinguishes dorsal (sensorimotor integration) and ventral (semantic processing) pathways [3]. Structural connectivity is a key determinant of language recovery. White matter integrity is particularly important because it supports communication between distributed language regions. The arcuate fasciculus connects frontal and temporal regions, and damage here predicts unfavourable outcomes. Disruption of structural connectivity often explains aphasic deficits more accurately than cortical lesion location alone, further supporting the network-based model of language [4]. Despite these advances, important methodological limitations remain. Resting-state fMRI captures connectivity at a single time point, whereas post-stroke recovery is a dynamic process unfolding over months. Consequently, cross-sectional studies cannot fully characterize longitudinal network reorganization. Longitudinal approaches are needed; without them, causal inferences remain elusive [5].

Emerging evidence suggests that contralesional recruitment depends on the degree of structural reserve [5,7]. Patients with preserved left-hemisphere networks generally benefit from restoration of ipsilesional function, whereas those with extensive structural damage may rely on compensatory right-hemisphere activation. This framework provides a mechanistic explanation for the apparently conflicting vicariation and interhemispheric competition models.

4. DEFINITION AND BIOLOGICAL BASIS OF NEUROPLASTICITY

Neuroplasticity refers to the brain's capacity to reorganise after injury. Two principal mechanisms underlie neuroplasticity: axonal sprouting—surviving neurons extend new connections—and synaptic reorganisation, whereby activity strengthens synapses through long-term potentiation-like processes [8]**.** Reactive plasticity occurs immediately after injury, whereas experience-dependent plasticity develops in response to rehabilitation and repeated behavioural training [3]. This distinction has important clinical implications because it helps define both the timing and intensity of therapeutic interventions.

Although axonal sprouting is well established in experimental models, direct evidence in humans remains limited, and much of the available knowledge is derived from animal studies [8]. Intensive language therapy increases perilesional activation, with 12 weeks of treatment reshaping cortical maps [9]. However, these findings were obtained in a relatively small cohort (n = 18) and require confirmation in larger prospective studies.

5. NEUROPLASTICITY MODELS IN POST-STROKE APHASIA

5.1. VICARIATION MODEL

This model posits that the right hemisphere compensates when left tissue is destroyed. Evidence derives from patients with extensive left lesions who still recover; functional imaging reveals right inferior frontal gyrus activation in such cases [7].

A recent theoretical and methodological review concluded that right-hemisphere recruitment correlates with better outcomes — but only in patients with severe left damage [5]. In those with mild damage, the same pattern predicts worse recovery. This nuance underscores that the same imaging finding carries opposite implications depending on lesion burden.

Critics raise two objections: right-hemisphere activity may be epiphenomenal (reflecting disinhibition rather than genuine compensation), and stimulating the right hemisphere in the wrong patient may exacerbate deficits**.** These concerns may be addressed by incorporating structural reserve into patient stratification [7].

5.2. INTERHEMISPHERIC COMPETITION MODEL

This model begins from a different premise: after a left-hemisphere stroke, the right hemisphere becomes hyperactive and inhibits the left perilesional cortex via transcallosal pathways. Reducing right-hemisphere activity, the argument goes, allows the left side to recover.

A meta-analysis synthesising 18 trials involving 612 patients demonstrated that inhibitory rTMS over the right inferior frontal gyrus improved naming and repetition with a moderate effect size (SMD ≈ 0.7) [10]. Heterogeneity, however, was considerable; some patients improved markedly, others not at all.

A systematic review of 14 tDCS trials found that effects were smaller than those observed with rTMS [11]. Gains were modest when tDCS was combined with language therapy. The authors criticise most studies for small samples and short treatment durations — few trials exceeded 10 sessions — which may explain the limited effects. Stimulation parameters varied widely (anodal, cathodal, bihemispheric protocols), and standardisation is lacking.

Publication bias also plagues both rTMS and tDCS evidence: negative trials are harder to locate. Assessment of publication bias suggested that the true treatment effect may be smaller than currently reported [10].

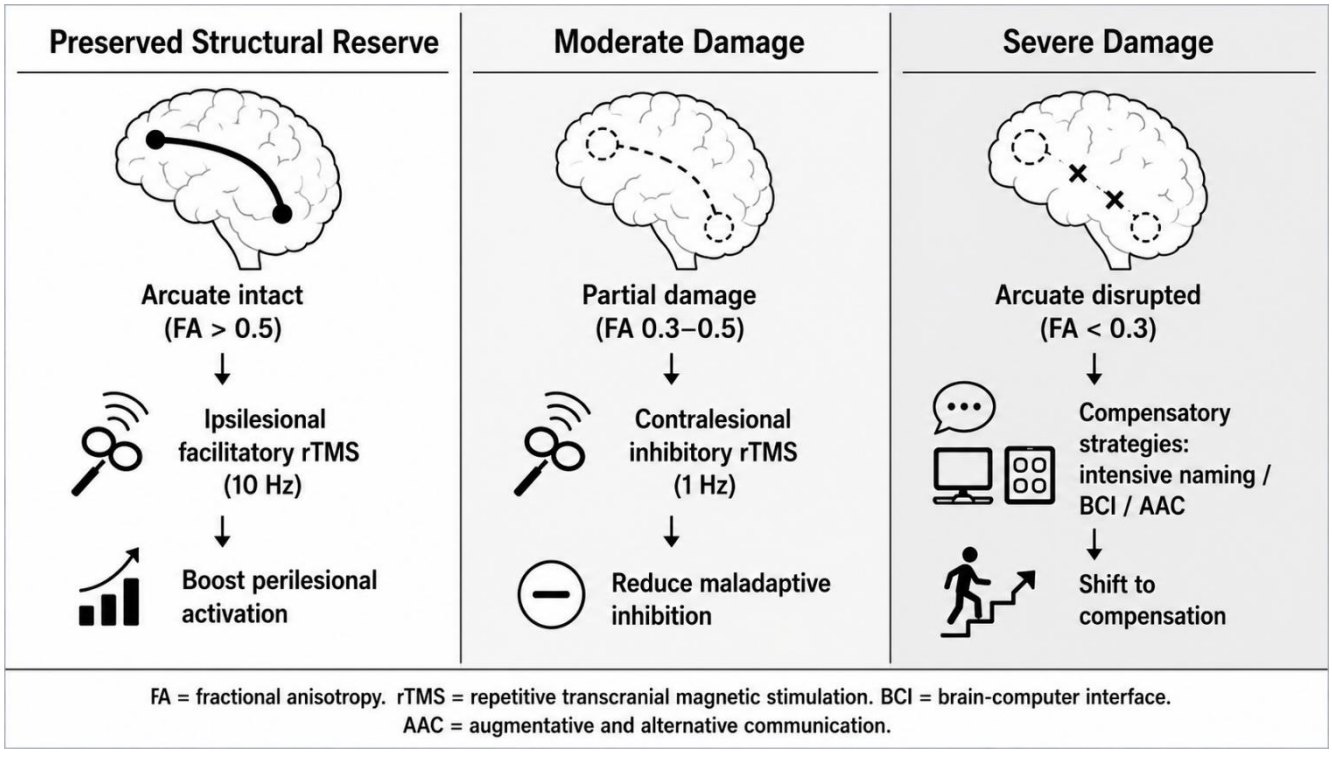


Figure 1: *Bimodal balance–recovery model for post-stroke aphasia. The optimal neuromodulation strategy depends on structural reserve of the left arcuate fasciculus measured by DTI.*

6. BIMODAL BALANCE-RECOVERY MODEL

The bimodal model resolves the contradiction by asserting

that the optimal pathway depends on a single factor: structural reserve. Preserved left networks favour ipsilesional facilitation; severely damaged left networks require right-hemisphere compensation. This framework remains hypothesis-generating, and alternative interpretations of imaging data exist [5, 7].

Table 1: Comparison of neuroplasticity models in post-stroke aphasia

Model	Core hypothesis	Clinical strategy	Best evidence	Key limitation
Vicariation	Right hemisphere compensates for left damage	Facilitate right hemisphere	Billot et al. [5]; Brancaccio et al. [7]	May be epiphenomenal; wrong patient may worsen
Interhemispheric competition	Right hyperactivity inhibits left recovery	Inhibit right hemisphere	Cheng et al. [10] (rTMS); Zettin et al. [11] (tDCS)	High heterogeneity; publication bias likely
Bimodal balance–recovery	Strategy depends on structural reserve	Choose based on DTI biomarkers	Billot et al. [5]; Brancaccio et al. [7]; Han et al. [6]	Not prospectively validated; cut-off values not standardised

A systematic review of 47 studies supported both recovery pathways depending on lesion characteristics and highlighted the value of diffusion tensor imaging for assessing structural reserve before treatment [5].

A proposed clinical algorithm includes [7] the following steps: (1) assessment of arcuate fasciculus integrity; (2) measurement of lesion volume; and (3) selection of the stimulation target—left for patients with an intact arcuate fasciculus and right for those with extensive disruption and large left-hemisphere lesions. This algorithm is testable, and prospective studies are ongoing.

Figure 1 illustrates the bimodal model's three patient-specific pathways, and Table 1 compares the three models side by side.

As Table 1 shows, the bimodal model offers the most flexible, patient-specific approach. Its main weakness is the absence of prospective validation.

Cut-off values for "preserved" versus "compromised" reserve remain unstandardised; different MRI scanners and sequences yield different DTI metrics.

6.1. TOWARDS STANDARISED BIOMARKERS FOR CLINICAL DECISION-MAKING

The bimodal model's clinical utility depends on measurable biomarkers. DTI currently offers the most promising avenue, with two metrics receiving greatest attention: fractional anisotropy (FA) — reflecting white matter coherence — and mean diffusivity (MD), which captures overall diffusion magnitude irrespective of direction.

In healthy individuals, left arcuate fasciculus FA typically ranges from 0.45 to 0.55. Lower FA predicts poorer naming outcomes at six months. However, the critical threshold separating "preserved" from "compromised" reserve remains debated. A three-tier classification has been proposed based on fractional anisotropy values: preserved (FA > 0.5), moderate damage (FA 0.3–0.5), and severe damage (FA < 0.3) [6]. Patients in the moderate category derived greatest benefit from rTMS; those with preserved FA improved with behavioural therapy alone; those with severe damage required contralesional compensatory approaches.

Other groups adopt different thresholds. Another approach defines significant damage as FA < 0.7 relative to the contralesional hemisphere [7]. Lesion volume has also been proposed as a practical surrogate when DTI is unavailable [5]: left-hemisphere lesions exceeding 30 cm^3 typically preclude ipsilesional facilitation. Combining DTI and volumetric data improves accuracy; a composite score integrating arcuate FA and lesion volume explained 62 % of variance in language outcomes at 12 months in one prospective study [5].

Functional biomarkers such as resting-state fMRI add another layer. Reduced interhemispheric functional connectivity between left and right inferior frontal gyri predicts better response to contralesional inhibitory rTMS; paradoxically, preserved connectivity favours ipsilesional facilitation. Yet fMRI protocols vary substantially across centres. Until standardisation improves, DTI remains the more clinically accessible and reproducible biomarker.

The absence of universally accepted cut-offs remains a barrier to clinical implementation. Different scanners (3T vs 1.5T), acquisition sequences, and post-processing pipelines produce non-identical FA values. A patient classified as "moderate" in one system might be "severe" in another. Multicentre trials with harmonised DTI protocols are urgently needed. DTI currently functions as a research tool rather than a routine clinical biomarker; its application outside specialised centres with standardised protocols is limited.

7. CLINICAL INTERVENTIONS SUPPORTING NEUROPLASTICITY

7.1. REPETITIVE TRANSCRANIAL MAGNETIC STIMULATION

The strongest current evidence for rTMS comes from a recent sham-controlled meta-analysis [10], which included only sham-controlled trials. Naming improved by 0.7 standard deviations above sham, and gains persisted at 1–3 months of follow-up. The optimal frequency (1, 10, or 20 Hz) remains unclear.

7.2. TRANSCRANIAL DIRECT CURRENT STIMULATION

A systematic review of 14 tDCS trials found that effects were smaller than for rTMS [11]. The authors criticise most studies for small samples and short treatment duration — few exceeded 10 sessions — which may explain modest results. Longer protocols may prove more effective.

7.3. INTENSIVE SPEECH LANGUAGE THERAPY

A national survey of Intensive Comprehensive Aphasia Programmes (ICAPs) in the UK demonstrated that 4–6 weeks of daily therapy yielded clinically meaningful gains, even in patients 2 years post-stroke [13]. Intensity mattered more than total duration; less than 10 hours per week produced minimal effects.

Real-world data from 448 patients showed a responder rate of 59 % [14]. Patients with severe aphasia showed the largest absolute gains but also the highest dropout rate — a sobering real-world finding that tempers optimism from controlled trials.

7.4. BRAIN-COMPUTER INTERFACES

A pilot study involving 12 patients with chronic aphasia evaluated an EEG-based brain-computer interface (BCI) [15]; participants learned to control a speller device, and language production improved on both trained and untrained items. Generalisation is promising, but the sample size is tiny.

A study using a visual P300 brain-computer interface in eight patients found that two could not operate the device at all [16]. Among the six who could, gains were modest. The authors conclude that BCI is not yet ready for clinical use and call for more research on user characteristics.

7.5. WHITE MATTER AND NETWORK PLASTICITY

A recent review demonstrated that arcuate fasciculus integrity predicts response to rTMS [6]. Patients with moderate damage respond best; those with no damage do not require rTMS; those with complete damage do not benefit — a pattern that directly supports the bimodal model.

A systematic review identified high-dose repetition, functional communication tasks, and psychosocial support as key components of ICAPs [12]. No single element sufficed alone, underscoring the value of multimodal rehabilitation.

7.6. SYNTHESIS AND CLINICAL PERSPECTIVE

Recent evidence similarly suggests that patient-specific factors outweigh universal recovery mechanisms [17]. Structural reserve determines the optimal approach — yet the bimodal model should be viewed as a conceptual framework rather than a proven clinical standard. Ongoing trials and future meta-analyses may refine or challenge its predictions.

The model has three strengths: it reconciles contradictory findings from rTMS and tDCS trials; it uses measurable biomarkers (DTI of the arcuate fasciculus); and it generates testable predictions. Its limitations are equally clear: prospective validation is lacking, cut-offs remain unstandardised, and most evidence derives from high-income countries with advanced MRI facilities; generalisability to other settings is unknown.

A further layer of complexity is clinical heterogeneity. Most neuromodulation studies target naming and repetition deficits, which respond more consistently to rTMS. Global aphasia with severe comprehension impairment shows more modest gains. Thus, the proposed framework is most applicable to non-fluent and anomic aphasias; its utility for severe global or mixed aphasias requires separate investigation. Cognitive impairments in young stroke survivors, including executive dysfunction and attention deficits, may further influence rehabilitation outcomes and should be considered in clinical decision-making [18].

Time-dependent recovery mechanisms also deserve attention. In the acute and early subacute phase (first 3 months), spontaneous recovery and reactive plasticity predominate; right-hemisphere recruitment during this period may reflect genuine compensation. In the chronic phase (≥6 months), maladaptive interhemispheric inhibition becomes more prominent, and inhibitory rTMS protocols show stronger evidence. Accordingly, the framework is most directly applicable to chronic aphasia; extending it to acute or subacute patients requires caution, as spontaneous recovery confounds treatment effects.

7.7. WHY CURRENT EVIDENCE REMAINS INSUFFICIENT FOR UNIVERSAL CLINICAL IMPLEMENTATION

Despite growing support for the bimodal model, several barriers remain. First, no large-scale prospective trial has tested biomarker-driven neuromodulation in aphasia. Second, DTI protocols vary substantially across centres, making FA thresholds non-transferable. Third, most studies exclude patients with severe aphasia or significant comorbidities, limiting generalisability to real-world populations. Until these gaps are addressed through multicentre harmonised protocols, clinical judgement must accompany biomarker-based stratification.

7.8. A PROPOSED CLINICAL FRAMEWORK FOR PATIENT STRATIFICATION

Translating the bimodal balance–recovery model into practice requires a structured approach. The following framework is hypothesis-generating and requires

prospective validation; it should not be interpreted as a clinical guideline. We propose it for patients with chronic post-stroke aphasia (≥6 months from onset).

Step 1: Assess structural reserve. Obtain DTI of the arcuate fasciculus on a 3T scanner if available. Measure FA in the left arcuate fasciculus: FA > 0.5 = preserved reserve; FA 0.3–0.5 = moderate damage; FA < 0.3 = severe damage. If DTI is unavailable, use lesion volume on T1-weighted MRI as a proxy: lesions <30 cm^3 suggest preserved reserve; lesions >30 cm^3 suggest compromised reserve.

Step 2: Select neuromodulation target based on classification.

Preserved reserve (FA > 0.5 or small lesion): ipsilesional facilitatory rTMS (10 Hz over left perilesional cortex, 10–20 sessions) combined with speech-language therapy — the goal is to boost perilesional activation.

Moderate damage (FA 0.3–0.5 or lesion 30–50 cm^3): contralesional inhibitory rTMS (1 Hz over right inferior frontal gyrus, 1200 pulses/session, 10–20 sessions) plus speech-language therapy — this reduces maladaptive interhemispheric inhibition, allowing residual left networks to recover.

Severe damage (FA < 0.3 or lesion >50 cm^3): ipsilesional facilitation is unlikely to succeed; contralesional inhibitory rTMS may still offer modest benefits, but the primary focus should shift to compensatory strategies: intensive naming therapy, augmentative communication devices, or BCI training in selected cases.

Step 3: Combine neuromodulation with intensive behavioural therapy. ICAP evidence shows that meaningful gains require a minimum intensity of 10 hours per week for 4–6 weeks. Neuromodulation without concurrent language therapy rarely yields lasting improvements. Task-oriented naming and repetition exercises appear most responsive to rTMS augmentation.

Step 4: Reassess after 20 sessions. If naming improves by ≥20 % on standardised testing, continue the same protocol for an additional 10–20 sessions. If no improvement occurs, consider switching to the alternative target or increasing behavioural therapy intensity.

The session numbers and thresholds are illustrative, drawn from typical protocols [10], and should be adapted to individual response and local resources.

Limitations of this framework must be acknowledged: it has not been prospectively validated in a multicentre trial; the proposed FA thresholds derive from single-centre studies [6] and require replication. Until international consensus emerges, clinical judgement should guide final decisions. Nonetheless, the framework provides a transparent, evidence-based starting point adaptable as new data become available.

8. CONCLUSIONS

Post-stroke language recovery operates at multiple levels: molecular plasticity enables synaptic strengthening; structural remodelling maintains network integrity; functional reorganisation redistributes language processing; experience-dependent adaptation drives therapeutic gains. Modern rehabilitation should follow three principles: (1) measure structural reserve before treatment; (2) choose neuromodulation target based on reserve; (3) combine neuromodulation with intensive task-oriented therapy. The bimodal balance–recovery model provides a roadmap; clinical trials testing this roadmap are the next step.

9. CONFLICT OF INTERESTS

The authors have no conflict of interest to declare. The authors declared that this study has received no financial support.

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